



Laboratorium
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Laporan Akhir

Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-Multivendor

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2026

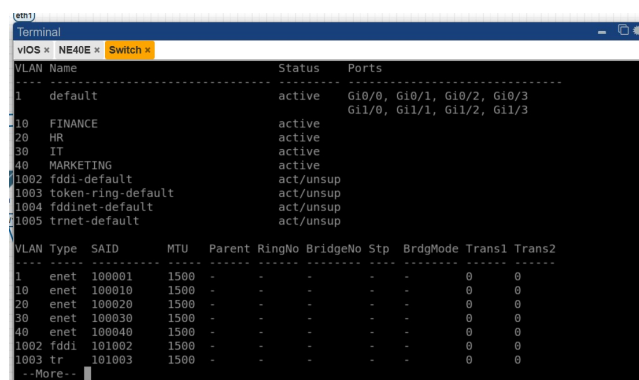
1 Langkah-Langkah Percobaan

Bagian ini menjelaskan secara rinci prosedur yang dilakukan selama praktikum, mulai dari persiapan alat hingga pelaksanaan setiap percobaan.

1.1 Praktikum 1: Konfigurasi Cisco Switch

Langkah Percobaan:

1. Masuk ke terminal Cisco Switch
2. Membuat Vlan
3. Pastikan Vlan sudah dibuat dan statusnya aktif

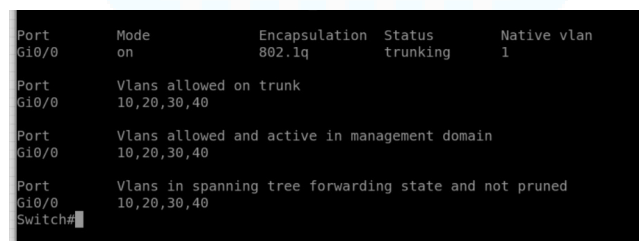


VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	FINANCE	active	
20	HR	active	
30	IT	active	
40	MARKETING	active	
1002	fdi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fdinet-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
40	enet	100040	1500	-	-	-	-	-	0	0
1002	fdi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0

Gambar 1: show vlan brief

4. Konfigurasi Access port untuk setiap vlan
5. Konfigurasi Trunk ke Cisco Router



Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gi0/0	10,20,30,40

Port	Vlans allowed and active in management domain
Gi0/0	10,20,30,40

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	10,20,30,40

Gambar 2: show interfaces trunk

1.2 Praktikum 2: Konfigurasi Cisco Router

Langkah Percobaan:

1. Masuk ke Cisco Router
2. Konfigurasi interface Trunk ke Switch
3. Konfigurasi Subinterface VLAN 10
4. Konfigurasi Subinterface VLAN 20

5. Konfigurasi Subinterface VLAN 30
6. Konfigurasi Subinterface VLAN 40
7. Konfigurasi DHCP-Server untuk VLAN 10
8. Konfigurasi DHCP-Server untuk VLAN 20
9. Konfigurasi IP Statis pada VPCS untuk VLAN 30 dan VLAN 40
10. Verifikasi Cisco Router

```

vcs> NE40E > Switch > IT > MARKETING > HR > FINANCE >
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#exit
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#exit
Router#show ip
*Jun  6 05:49:00.798: %SYS-5-CONFIG I: Configured from console by consolein
% Ambiguous command: "show ip in"
Router#show ip interface brief
Interface      IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0      unassigned      YES unset  administratively down down
GigabitEthernet0/1      unassigned      YES unset  administratively down down
GigabitEthernet0/2      unassigned      YES unset  up                  up
GigabitEthernet0/2.10   192.168.10.1    YES manual  up                  up
GigabitEthernet0/2.20   192.168.20.1    YES manual  up                  up
GigabitEthernet0/2.30   192.168.30.1    YES manual  up                  up
GigabitEthernet0/2.40   192.168.40.1    YES manual  up                  up
GigabitEthernet0/3      unassigned      YES unset  administratively down down
Router#

```

Gambar 3: show ip interface brief

```

vcs> NE40E > Switch > IT > MARKETING > HR > FINANCE >
GigabitEthernet0/2.20   192.168.20.1    YES manual  up                  up
GigabitEthernet0/2.30   192.168.30.1    YES manual  up                  up
GigabitEthernet0/2.40   192.168.40.1    YES manual  up                  up
GigabitEthernet0/3      unassigned      YES unset  administratively down down
Router#show ip dhcp pool
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range   Leased addresses
    192.168.10.12   192.168.10.1 - 192.168.10.254   1
Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range   Leased addresses
    192.168.20.12   192.168.20.1 - 192.168.20.254   1
Router#

```

Gambar 4: show ip dhcp pool

11. Pengujian dari VPCS

```

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.2c Jun 07 2026 05:48 AM    Automatic
192.168.20.11    0100.5079.6668.2d Jun 07 2026 05:46 AM    Automatic
Router#

```

Gambar 5: show ip dhcp binding

```
Terminal
VPCS> ip dhcp
300
can't find dhcp server

VPCS> ip dhcp
30000 IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86284, 86400/43200/75600
MAC        : 00:50:79:66:68:2c
I/P/PORT    : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> █
```

Gambar 6: ip dhcp dan show ip dari VLAN 10 (FINANCE)

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.578 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.034 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.688 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.273 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=4.112 ms

VPCS> █
```

Gambar 7: Ping Gateway 192.168.10.1 dari VLAN 10

```
Terminal
VPCS> ip dhcp
30000 IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86129, 86400/43200/75600
MAC        : 00:50:79:66:68:2d
I/P/PORT    : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.507 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.061 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=3.022 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=4.233 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=4.782 ms

VPCS> █
```

Gambar 8: ip dhcp, show ip, dan Ping Gateway 192.168.20.1 dari VLAN 20 (HR)

```
VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.260 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.199 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=5.010 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=5.346 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.325 ms

VPCS> █
```

Gambar 9: Ping Gateway 192.168.30.1 dari VLAN 30 (IT)

```
VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.166 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.387 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=4.285 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=4.865 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=4.191 ms

VPCS> █
```

Gambar 10: Ping Gateway 192.168.40.1 dari VLAN 40 (MARKETING)

```

VPCS> ping 192.168.20.11
Pinging 192.168.20.11: icmp_seq=4 ttl=255 time=4.448 ms
Pinging 192.168.20.11: icmp_seq=5 ttl=255 time=3.843 ms (ICMP type:3, code:1, Destination host unreachable)
VPCS> ping 192.168.20.11
64 bytes from 192.168.20.11: icmp_seq=1 ttl=63 time=11.444 ms
64 bytes from 192.168.20.11: icmp_seq=2 ttl=63 time=0.351 ms
64 bytes from 192.168.20.11: icmp_seq=3 ttl=63 time=0.029 ms
64 bytes from 192.168.20.11: icmp_seq=4 ttl=63 time=0.128 ms
64 bytes from 192.168.20.11: icmp_seq=5 ttl=63 time=5.692 ms
VPCS> ping 192.168.30.10
64 bytes from 192.168.30.10: icmp_seq=1 ttl=63 time=10.884 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=63 time=4.367 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=63 time=0.392 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=63 time=4.252 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=63 time=0.587 ms
VPCS> ping 192.168.40.10
Bad command: "192.168.40.10". Use ? for help.
VPCS> ping 192.168.40.10
64 bytes from 192.168.40.10: icmp_seq=1 ttl=63 time=10.180 ms
64 bytes from 192.168.40.10: icmp_seq=2 ttl=63 time=4.165 ms
64 bytes from 192.168.40.10: icmp_seq=3 ttl=63 time=5.664 ms
64 bytes from 192.168.40.10: icmp_seq=4 ttl=63 time=4.628 ms
64 bytes from 192.168.40.10: icmp_seq=5 ttl=63 time=4.733 ms
VPCS>

```

Gambar 11: Uji ping antar-Vlan dari VLAN 10

```

Jun 06 06:00:10:959: %SYS-5-CONFIG-I- Configured from console by console binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.2c Jun 07 2026 05:48 AM   Automatic
192.168.20.11    0100.5079.6668.2d Jun 07 2026 05:46 AM   Automatic
Router#

```

Gambar 12: show ip dhcp binding lagi

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

Langkah Percobaan:

1. Konfigurasi Lan Huawei

```

VPCS> NSMKE Switch> MARKETING> IT > HR > FINANCE>
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
monitor[2952]: [SVRPTIME(2952) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
location[2957]: [SVRPTIME(2957) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
location[2959]: [SVRPTIME(2959) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:06 UTC):
boot[2958]: [SVRPTIME(2958) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
start fail first time.iret=0
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:09 UTC):
location[3861]: [SVRPTIME(3861) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date
Broadcast message from systemd-journal@debian (Sat 2026-06-06 06:06:09 UTC):
location[3883]: [SVRPTIME(3883) VRP_InitSvrpTimeSharedDate-417] Not OSN SVRP, do not need to init svrp time shared date

```

Gambar 13: Menunggu Router Huawei

2. Konfigurasi IP VPC Lan Huawei

```

Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:4b
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
VPCS>

```

Gambar 14: show ip vpc lan Huawei

```

VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=15.015 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.436 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.871 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.959 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.008 ms
VPCS>

```

Gambar 15: ping gateway huawei (192.168.50.1)

3. Konfigurasi IP Link Cisco <-> Huawei Link 1
4. Konfigurasi IP Link Cisco <-> Huawei Link 2
5. Konfigurasi IP Link Huawei <-> Mikrotik
6. Konfigurasi IP Link Cisco <-> Mikrotik
7. Verifikasi Interface

```

% Ambiguous command: "show ip i"
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.10.11.1	YES	manual	up	up
GigabitEthernet0/1	10.10.30.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/2.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/3	10.10.10.1	YES	manual	up	up

Gambar 16: show ip interface brief (Cisco)

```

[~HUAWEI]display ip interface brief

```

*down: administratively down
 !down: FIB overload down
 ^down: standby
 (L): loopback
 (s): spoofing
 (d): Dampening Suppressed
 (E): E-Trunk down

The number of interface that is UP in Physical is 23
 The number of interface that is DOWN in Physical is 0
 The number of interface that is UP in Protocol is 16
 The number of interface that is DOWN in Protocol is 7

Interface	IP Address/Mask	Physical	Protocol	VPN
Ethernet1/0/0	10.10.10.2/30	up	up	--
Ethernet1/0/0.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/1	unassigned	up	down	--
Ethernet1/0/1.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/2	192.168.50.1/24	up	up	--
Ethernet1/0/2.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/3	10.10.20.1/30	up	up	--
Ethernet1/0/3.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/4	10.10.11.2/30	up	up	--
Ethernet1/0/4.4094	128.1.138.137/16	up	up	l3vpn

Gambar 17: display ip interface brief (Huawei)

```

[admin@MikroTik] > /ip address print

```

Flags: X - disabled, I - invalid, D - dynamic

#	ADDRESS	NETWORK	INTERFACE
0	;; TO-HUAWEI 10.10.20.2/30	10.10.20.0	ether1
1	;; TO-CISCO 10.10.30.2/30	10.10.30.0	ether2

Gambar 18: /ip address print (Mikrotik)

8. Verifikasi ping Antar Router

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

Gambar 19: Ping dari Router Cisco

```

Terminal
vIOS x NE40E x Switch x MARKETING x IT x HR x FINANCE x VPC x Mikrotik x

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 3/3/3 ms

[~HUAWEI]

```

Gambar 20: Ping dari Router Huawei

```

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=3 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/3 ms

[~HUAWEI]

```

Gambar 21: Ping dari Router Huawei (part 2)

```

Terminal
HOS > NE40E > Switch > MARKETING > IT > HR > FINANCE > VPC > Mikrotik >

(admin@Mikrotik) > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME STATUS
0 10.10.20.1                            56 255 3ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 2ms
4 10.10.20.1                            56 255 1ms
5 10.10.20.1                            56 255 1ms
6 10.10.20.1                            56 255 1ms
7 10.10.20.1                            56 255 1ms
8 10.10.20.1                            56 255 1ms
sent=9 received=9 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms

(admin@Mikrotik) > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME STATUS
0 10.10.30.1                            56 255 1ms
1 10.10.30.1                            56 255 2ms
2 10.10.30.1                            56 255 1ms
3 10.10.30.1                            56 255 2ms
4 10.10.30.1                            56 255 2ms
5 10.10.30.1                            56 255 2ms
6 10.10.30.1                            56 255 1ms
sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

(admin@Mikrotik) >

```

Gambar 22: Ping dari Router Mikrotik

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

Langkah Percobaan:

1. Konsep Routing OSPF pada Topologi

Link	Cost	Fungsi
Cisco ↔ Huawei Link 1	10	Jalur utama
Cisco ↔ Huawei Link 2	10	Jalur utama
Cisco ↔ MikroTik	100	Jalur backup
Huawei ↔ MikroTik	100	Jalur backup

Gambar 23: Konsep Routing

2. Network yang diiiklankan ke OSPF

Cisco Router	
Network	Keterangan
10.10.10.0/30	Link Cisco-Huawei 1
10.10.11.0/30	Link Cisco-Huawei 2
10.10.30.0/30	Link Cisco-MikroTik
192.168.10.0/24	VLAN 10 Finance
192.168.20.0/24	VLAN 20 HR
192.168.30.0/24	VLAN 30 IT
192.168.40.0/24	VLAN 40 Marketing

Huawei Router	
Network	Keterangan
10.10.10.0/30	Link Huawei-Cisco 1
10.10.11.0/30	Link Huawei-Cisco 2
10.10.20.0/30	Link Huawei-MikroTik
192.168.50.0/24	LAN Huawei

Mikrotik	
Network	Keterangan
10.10.20.0/30	Link MikroTik-Huawei
10.10.30.0/30	Link MikroTik-Cisco

Gambar 24: Network ke OSPF

3. Konfigurasi OSPF pada Cisco Router

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

Router#

```

Gambar 25: show ip protocols

4. Konfigurasi OSPF pada Huawei Router

```

Terminal
VLOS > [HUAWEI-Ethernet1/0/3]quit
[~HUAWEI]commit
[~HUAWEI]display ospf peer

(N) Indicates MADJ neighbor

  OSPF Process 1 with Router ID 2.2.2.2
    Neighbors
  Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
    Router ID: 1.1.1.1      Address: 10.10.10.1
    State: Full           Mode: Nbr is Slave   Priority: 1
    DR: 10.10.10.1       BDR: 10.10.10.2     MTU: 1500
    Dead timer due in 30 sec
    Retrans timer interval: 5
    Neighbor is up for 00h01m50s
    Neighbor Up Time : 2026-06-06 06:29:29
    Authentication Sequence: [ 0 ]

  Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
    Router ID: 1.1.1.1      Address: 10.10.11.1
    State: Full           Mode: Nbr is Slave   Priority: 1
    DR: 10.10.11.1       BDR: 10.10.11.2     MTU: 1500
    Dead timer due in 37 sec
    Retrans timer interval: 5
    Neighbor is up for 00h01m50s
    Neighbor Up Time : 2026-06-06 06:29:29
    Authentication Sequence: [ 0 ]

[~HUAWEI]

```

Gambar 26: display ospf peer

```

Terminal
VLOS > [HUAWEI-Ethernet1/0/3]quit
[~HUAWEI]commit
[~HUAWEI]display ip routing-table protocol ospf

Route flags: R - relay, D - download to fib, ? - to vpn-instance, B - black hole route
.....
public Routing Table : OSPF          Routes : 14
OSPF routing table status : <Active>
  Destinations : 9          Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop        Interface
-----
10.10.30.0/30       OSPF     10     110      D  10.10.11.1  Ethernet1/0/4
10.10.11.0/30       OSPF     10     110      D  10.10.10.1  Ethernet1/0/0
192.168.10.0/24      OSPF     10     11        D  10.10.11.1  Ethernet1/0/4
192.168.20.0/24      OSPF     10     11        D  10.10.10.1  Ethernet1/0/0
192.168.30.0/24      OSPF     10     11        D  10.10.11.1  Ethernet1/0/4
192.168.40.0/24      OSPF     10     11        D  10.10.10.1  Ethernet1/0/0
192.168.50.0/24      OSPF     10     11        D  10.10.11.1  Ethernet1/0/4
192.168.60.0/24      OSPF     10     11        D  10.10.10.1  Ethernet1/0/0

OSPF routing table status : <Inactive>
  Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre    Cost    Flags NextHop        Interface
-----
10.10.10.0/30       OSPF     10     10        D  10.10.10.2  Ethernet1/0/0
10.10.11.0/30       OSPF     10     10        D  10.10.11.2  Ethernet1/0/4
10.10.20.0/30       OSPF     10     100       D  10.10.20.1  Ethernet1/0/3
192.168.50.0/24     OSPF     10     1         D  192.168.50.1 Ethernet1/0/2

[~HUAWEI]

```

Gambar 27: display ip routing-table protocol ospf

```

Terminal
vIOS > NEAGE > Switch > MARKETING > IT > HR > FINANCE > VPC > Mikrotik >

Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.2 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAMEI]

```

Gambar 28: display ospf routing

5. Konfigurasi OSPF pada Mikrotik

```

[admin@mikrotik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=0.0.0.0 state="ExStart" state-changes=3 ls-retransmits=0 ls-requests=0 db-summaries=0
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=5s
[admin@mikrotik] >

```

Gambar 29: /routing ospf neighbor print

```

[admin@mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
  10.10.30.1
1 Ado 10.10.11.0/30 10.10.20.1 110
  10.10.30.1
2 Ado 192.168.10.0/24 10.10.30.1 110
  10.10.30.1
3 Ado 192.168.20.0/24 10.10.30.1 110
  10.10.30.1
4 Ado 192.168.30.0/24 10.10.30.1 110
  10.10.30.1
5 Ado 192.168.40.0/24 10.10.30.1 110
  10.10.30.1
6 Ado 192.168.50.0/24 10.10.20.1 110
[admin@mikrotik] >

```

Gambar 30: /ip route print where ospf

6. Verifikasi OSPF Neighbor

```

Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:36 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:38 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:34 10.10.10.2 GigabitEthernet0/3
Router#

```

Gambar 31: Verifikasi OSPF Neighbor Cisco Router

```

Terminal
vIOS > NEAGE > Switch > MARKETING > IT > HR > FINANCE > VPC > Mikrotik >

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/3)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 39 sec
Retrans timer interval: 5
Neighbor is up for 00h05m27s
Neighbor Up Time : 2026-06-06 06:29:29
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h05m27s
Neighbor Up Time : 2026-06-06 06:33:45
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00h05m27s
Neighbor Up Time : 2026-06-06 06:29:29
Authentication Sequence: [ 0 ]
[~HUAMEI]

```

Gambar 32: Verifikasi OSPF Neighbor Huawei Router

```

admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=1m40s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=1m48s
admin@Mikrotik] >

```

Gambar 33: Verifikasi OSPF Neighbor Mikrotik

7. Verifikasi Routing Table

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:18 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:06:18 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:06:18 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#

```

Gambar 34: Verifikasi Routing Table Cisco Router menuju Lan Huawei

```

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:55, GigabitEthernet0/0
      [110/110] via 10.10.10.2, 00:04:55, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:40, GigabitEthernet0/0
      [110/11] via 10.10.10.2, 00:06:40, GigabitEthernet0/3
Router#

```

Gambar 35: cek semua route ospf (Cisco)

```

Terminal
VPCS > HUAWEI Switch > MARKETIND > IT > HR > FINANCE > VPC > Mikrotik >
Route flags: B - relay, D - download to fib, I - to vpn-instance, B - black hole route
Public Routing table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
10.10.11.0/30 OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.50.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/2
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]

```

Gambar 36: cek semua route ospf (HUAWEI)

```

admin@Mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
       - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110
admin@Mikrotik] >

```

Gambar 37: cek semua route ospf (Mikrotik)

8. Verifikasi Koneksi End-To-End

```
Terminal
VPCS > ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=17.595 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=8.206 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=10.011 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=5.909 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=6.551 ms
VPCS>
```

Gambar 38: Ping Gateway Huawei (192.168.50.1) dari VLAN 10 (FINANCE)

```
Terminal
VPCS > ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=8.326 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=9.276 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=9.645 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=7.116 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=9.282 ms
VPCS>
```

Gambar 39: Ping Gateway Huawei (192.168.50.1) dari VLAN 20 (HR)

```
Terminal
VPCS > ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=7.291 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=10.006 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=6.695 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=6.309 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=5.930 ms
VPCS>
```

Gambar 40: Ping Gateway Huawei (192.168.50.1) dari VLAN 30 (IT)

```
Terminal
VPCS > ping 192.168.50.10
64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=7.259 ms
64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=7.798 ms
64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=6.738 ms
64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=5.976 ms
64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=6.645 ms
VPCS>
```

Gambar 41: Ping Gateway Huawei (192.168.50.1) dari VLAN 40 (MARKETING)

```
Terminal
VPCS > ping 192.168.10.11
64 bytes from 192.168.10.11: icmp_seq=1 ttl=62 time=5.763 ms
64 bytes from 192.168.10.11: icmp_seq=2 ttl=62 time=7.343 ms
64 bytes from 192.168.10.11: icmp_seq=3 ttl=62 time=6.581 ms
64 bytes from 192.168.10.11: icmp_seq=4 ttl=62 time=7.012 ms
64 bytes from 192.168.10.11: icmp_seq=5 ttl=62 time=7.432 ms
VPCS> ping 192.168.20.11
64 bytes from 192.168.20.11: icmp_seq=1 ttl=62 time=6.548 ms
64 bytes from 192.168.20.11: icmp_seq=2 ttl=62 time=6.026 ms
64 bytes from 192.168.20.11: icmp_seq=3 ttl=62 time=7.515 ms
64 bytes from 192.168.20.11: icmp_seq=4 ttl=62 time=6.388 ms
64 bytes from 192.168.20.11: icmp_seq=5 ttl=62 time=6.237 ms
VPCS> ping 192.168.30.10
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=7.078 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=6.539 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=6.228 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=9.168 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=8.209 ms
```

Gambar 42: Ping dari PC Lan Huawei ke IP VLAN Cisco

```
VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=7.791 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=6.405 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=6.698 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=9.075 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=8.409 ms

VPCS>
```

Gambar 43: Ping dari PC Lan Huawei ke IP VLAN Cisco (part 2)

1.5 Praktikum 5: Pengujian Failover OSPF

Langkah Percobaan:

1. Kondisi Normal: Dua link Cisco-Huawei aktif

```
VPCS> trace 192.168.50.10
tracert to 192.168.50.10, 8 hops max
 1 192.168.50.10 0.001 ms
```

Gambar 44: trace dari pc vlan menuju lan huawei (Kondisi Normal)

2. Kondisi 1 link Cisco-Huawei mati

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 4.828 ms 6.104 ms 4.002 ms
 2 10.10.11.2 6.361 ms 5.135 ms 3.211 ms
 3 *192.168.50.10 11.337 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 45: trace dari pc vlan menuju lan huawei (Kondisi 1)

3. Kondisi Dua Link Cisco-Huawei Mati

```

VPCS > NEAR > Switch > MARKETING > IT > HR > FINANCE > VPC > Mikrotik >
GigabitEthernet0/3 10.10.10.1 YES manual up
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#
Jun 6 06:52:12.069: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Inter-
face down or detached
% Ambiguous command: "e"
Router(config)#
Jun 6 06:52:14.038: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
Jun 6 06:52:15.038: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config-if)#exit
Router#
Jun 6 06:52:17.331: %SYS-5-CONFIO-1: Configured from console by consoleconf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#shutdown
Router(config-if)#exit
Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual administratively down down
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual administratively down down
Router(config)#

```

Gambar 46: Kondisi Shutdown

4. Mengaktifkan Kembali Dua Link Cisco Huawei

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 5.967 ms 4.179 ms 4.396 ms
 2 10.10.11.2 7.091 ms 5.054 ms 6.015 ms
 3 *192.168.50.10 8.867 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 47: trace dari pc vlan menuju lan huawei (Kondisi Dua Link Aktif Kembali)

2 Analisis Hasil Percobaan

2.1 Praktikum 1: Konfigurasi Cisco Switch

Empat VLAN berhasil dibuat pada Cisco Switch, yaitu VLAN 10 (FINANCE), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (MARKETING), dengan seluruhnya berstatus *active* pada output `show vlan brief` (Gambar 1). Setiap port dikonfigurasi sebagai access port sesuai VLAN-nya, sedangkan port GigabitEthernet0/0 dikonfigurasi sebagai trunk dengan encapsulation 802.1Q menuju Cisco Router. Output `show interfaces trunk` pada Gambar 2 mengonfirmasi bahwa keempat VLAN sudah diizinkan melintas dan aktif dalam management domain, yang menandakan segmentasi jaringan berjalan sesuai rancangan.

2.2 Praktikum 2: Konfigurasi Cisco Router

Cisco Router dikonfigurasi dengan metode *Router-on-a-Stick* menggunakan empat subinterface pada port fisik GigabitEthernet0/2, masing-masing dengan gateway 192.168.10.1/24, 192.168.20.1/24, 192.168.30.1/24, dan 192.168.40.1/24 untuk tiap VLAN. Keempat subinterface tercatat berstatus UP/UP pada `show ip interface brief` (Gambar 3). DHCP Server dikonfigurasi untuk VLAN 10 dan VLAN 20, dan berhasil mendistribusikan IP 192.168.10.11 serta 192.168.20.11 ke VPCS secara otomatis, sebagaimana terlihat pada `show ip dhcp binding` di Gambar 5 dan 12. Pengujian ping dari seluruh VPCS ke masing-masing gateway berhasil dengan latensi 3 hingga 5 ms, dan uji ping antar-VLAN pada Gambar 11 membuktikan inter-VLAN routing berfungsi dengan baik.

2.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

Tiga router dari vendor berbeda dihubungkan menggunakan empat link *point-to-point* dengan subnet /30, mencakup dua link Cisco-Huawei (10.10.10.0/30 dan 10.10.11.0/30), satu link Huawei-Mikrotik (10.10.20.0/30), dan satu link Cisco-Mikrotik (10.10.30.0/30). LAN Huawei menggunakan subnet 192.168.50.0/24. Verifikasi interface pada Gambar 16, 17, dan 18 menunjukkan seluruh port aktif di ketiga router. Pengujian ping antarrouter pada Gambar 19 hingga 22 menghasilkan konektivitas yang baik dengan zero packet loss pada sebagian besar pasangan, yang mengonfirmasi konektivitas fisik antarrouter sudah terbentuk sebelum OSPF dikonfigurasi.

2.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

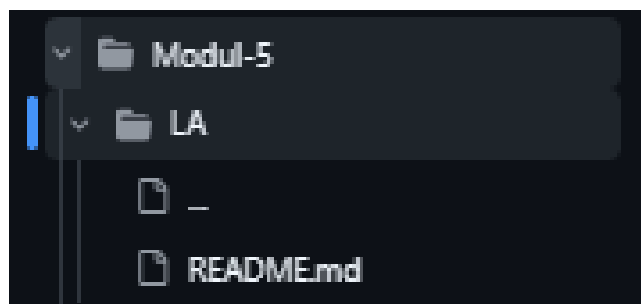
OSPF area 0 dikonfigurasi pada ketiga router dengan cost 10 untuk link Cisco-Huawei sebagai jalur utama dan cost 100 untuk link yang melibatkan Mikrotik sebagai jalur backup (Gambar 23). Adjacency OSPF berhasil terbentuk dengan status FULL pada semua pasangan router lintas vendor, sebagaimana terlihat pada Gambar 31, 32, dan 33. Routing table Cisco pada Gambar 34 memperlihatkan route menuju 192.168.50.0/24 muncul dengan dua next-hop sekaligus (metric 11), yang menandakan OSPF melakukan *equal-cost load balancing* secara otomatis pada dua link utama. Pengujian end-to-end pada Gambar 38 hingga 43 membuktikan konektivitas penuh antara VPCS di sisi Cisco dan PC di LAN Huawei dari kedua arah, dengan latensi 3 hingga 10 ms.

2.5 Praktikum 5: Pengujian Failover OSPF

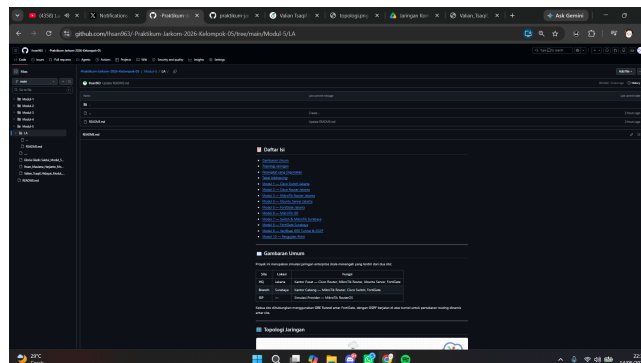
Pada kondisi normal, traceroute dari VPCS ke LAN Huawei menunjukkan paket tiba dalam satu hop dengan latensi sangat rendah (Gambar 44), sesuai ekspektasi karena OSPF memilih jalur cost 10. Saat satu link Cisco-Huawei dimatikan, OSPF berhasil melakukan konvergensi dan trafik tetap mengalir melalui link yang tersisa meski dengan sedikit gangguan sementara (Gambar 45). Ketika kedua link Cisco-Huawei dimatikan sekaligus (Gambar 46), trafik berhasil dialihkan secara otomatis melalui Mikrotik sebagai jalur backup dengan latensi meningkat menjadi 7 hingga 9 ms (Gambar 47), konsisten dengan cost 100 yang sudah dirancang. Setelah link diaktifkan kembali, OSPF langsung konvergen ulang dan memilih jalur utama tanpa perlu intervensi manual.

3 Hasil Tugas Modul

Untuk Tugas Modul telah di push di github kelompok kami dalam bentuk README.md, berikut merupakan bukti bahwa README telah di push:



Gambar 48: README



Gambar 49: Bukti tanpa Crop

4 Kesimpulan

Praktikum Modul 5 berhasil mengimplementasikan jaringan multivendor yang mengintegrasikan VLAN, Trunking, dan OSPF secara menyeluruh. Segmentasi jaringan dengan empat VLAN dan metode 802.1Q Trunking terbukti berjalan baik, begitu pula inter-VLAN routing menggunakan metode *Router-on-a-Stick* yang mampu melayani seluruh segmen dengan latensi rendah. DHCP Server pada Cisco Router juga berfungsi sebagaimana mestinya dalam mendistribusikan IP ke klien secara otomatis.

Pada sisi multivendor, OSPF area 0 berhasil menghubungkan Cisco, Huawei, dan Mikrotik dalam satu domain routing tanpa konfigurasi khusus per vendor. Adjacency Full terbentuk di semua pasangan router, dan desain cost yang membedakan jalur utama dan backup menghasilkan perilaku routing yang deterministik, termasuk *equal-cost load balancing* pada dua link Cisco-Huawei. Pengujian failover membuktikan bahwa OSPF mampu mengalihkan trafik ke jalur backup secara otomatis saat jalur utama gagal, dan kembali ke jalur utama tanpa intervensi saat link pulih. Praktikum ini secara keseluruhan menunjukkan bahwa pemilihan cost OSPF yang tepat dan desain redundansi yang baik adalah kunci keandalan sebuah jaringan.

5 Lampiran

5.1 Dokumentasi saat praktikum